

DEMO-102:2026

Illustrative Packaged Natural Mineral Water Specification (synthetic demo document, not an Indian Standard)

Second Revision - ICS 13.060.20

FOREWORD

This is a synthetic demonstration document created for the BISense prototype. It is not an Indian Standard, it has not been issued by the Bureau of Indian Standards, and all values in it are illustrative only.

Natural mineral water is distinguished from ordinary drinking water by its origin from an underground source and by its natural mineral content, which shall not be altered by treatment. This illustrative specification lays down requirements for such water.

1 SCOPE

1.1 This document prescribes the requirements and the methods of sampling and test for packaged natural mineral water.

1.2 Water that has been treated by reverse osmosis or to which minerals have been added is not covered by this document; see DEMO-101.

2 REFERENCES

The following documents contain provisions which, through reference in this text, constitute provisions of this document:

DEMO-101 Illustrative Packaged Drinking Water Specification

DEMO-900 Illustrative Rules for Rounding Off Numerical Values

DEMO-905 Illustrative Methods of Sampling and Microbiological Examination of Water

3 TERMINOLOGY

For the purpose of this document, the following definitions shall apply.

3.1 Natural Mineral Water - Water obtained directly from a natural or drilled underground source, protected from pollution, characterized by its content of certain mineral salts and their relative proportions, and packed close to the point of emergence of the source.

3.2 Source - The point of emergence of the underground water, such as a spring or a bore well.

3.3 Batch - All containers of the same capacity packed on one line in one shift from the same source.

4 REQUIREMENTS

4.1 Source Protection and Treatment

4.1.1 The source shall be protected against all risks of pollution by a protection zone.

4.1.2 The water shall not be subjected to any treatment other than separation of unstable elements by decantation or filtration, and the addition or removal of carbon dioxide.

4.1.3 Minerals shall not be added to the water.

4.1.4 The source should be monitored for flow rate and temperature every month.

4.2 Physical and Chemical Requirements

The water shall comply with the physical and chemical requirements given in Table 1 when tested by the methods referred to in column 5 of Table 1.

Table 1 Physical and Chemical Requirements for Packaged Natural Mineral Water

SI No.	Characteristic	Unit	Requirement	Method of Test
i	Colour, Max	Hazen units	2	Clause 8.1
ii	Odour	-	Agreeable	Clause 8.2
iii	Turbidity, Max	NTU	2	Clause 8.3
iv	Total dissolved solids	mg/l	150 to 700	Clause 8.4
v	pH value	-	6.5 to 8.5	Clause 8.5
vi	Nitrate (as NO ₃), Max	mg/l	50	Clause 8.6
vii	Fluoride (as F), Max	mg/l	1.0	Clause 8.6
viii	Chloride (as Cl), Max	mg/l	200	Clause 8.6
x	Calcium (as Ca), Max	mg/l	100	Clause 8.6
xi	Magnesium (as Mg), Max	mg/l	50	Clause 8.6

4.3 Microbiological Requirements

4.3.1 Coliform bacteria shall be absent in any 250 ml sample of the water.

4.3.2 Escherichia coli shall be absent in any 250 ml sample of the water.

4.3.3 The total viable colony count at 37 degrees C shall not exceed 20 per ml within 12 hours of packing.

5 PACKING

5.1 The water shall be packed at the source in clean, hermetically sealed food-grade containers.

5.2 Containers shall not be filled beyond 97 percent of their capacity.

6 MARKING

6.1 Each container shall be legibly and indelibly marked with the following information:

- Name of the product, namely "Natural Mineral Water";
- Name and location of the source;
- Name and address of the manufacturer;
- Batch number and date of packing;
- Net volume in ml or litres;
- The mineral composition in mg/l; and
- The words "Synthetic demo label" for this illustrative document.

6.2 The label shall not describe the water as having any medicinal property.

7 SAMPLING

7.1 Representative samples of the water shall be drawn as prescribed in DEMO-905.

7.2 The number of containers to be selected from a batch shall be as given in Table 2.

Table 2 Scale of Sampling

Batch Size (Containers)	Number of Containers to be Selected
Up to 5 000	6
5 001 to 25 000	9
25 001 and above	12

8 TEST METHODS

8.1 Colour - Colour shall be determined by visual comparison with platinum-cobalt standards.

8.2 Odour - Odour shall be assessed by at least two trained assessors at room temperature.

8.3 Turbidity - Turbidity shall be determined with a nephelometer calibrated with formazin standards.

8.4 Total Dissolved Solids - A filtered sample of 100 ml shall be evaporated at 180 degrees C and the residue weighed.

8.5 pH - pH shall be measured with a pH meter calibrated with two buffer solutions.

8.6 Mineral Content - Calcium, magnesium, nitrate, fluoride and chloride should be determined by ion chromatography or atomic absorption spectrometry.

8.7 Microbiological Examination - Coliform bacteria and Escherichia coli shall be tested by the membrane filtration method prescribed in DEMO-905.